Categories of AI: 1

Advanced: Practice 2

To distinguish between different levels of intelligence, researchers often divide artificial intelligence (AI) into three categories: narrow intelligence, general intelligence and super intelligence.

For each of the following examples of artificial intelligence, pick the most appropriate AI category. Drag the category into the correct row of the table below to label the example described.

Example	Category
A machine with intelligence that surpasses that of humans across all possible domains.	
An autonomous pizza delivery vehicle that could make its way to your home from the pizza shop, without human intervention, to deliver your pizza.	
A machine with the ability to understand, learn, and apply knowledge across a wide range of tasks at human-level proficiency.	
A computer system that can consistently beat the world number one chess player and has never itself been beaten.	

Items:

Super intelligence General intelligence Narrow intelligence

Quiz:
STEM SMART Computer Science Week 30

Social and ethical: data bias

Core: Practice 2

A comprehensive understanding of the social and ethical issues introduced by AI is essential to ensure the responsible development and deployment of these technologies.

An essential aspect to take into account involves examining potential **bias within the data** that is used for training machine learning models.

A university is developing an AI application that will predict the likelihood of a student graduating with a first class honours degree. Which of the following sets of training data would be **least likely** to produce a model that reveals bias?

- A set of data collected by a national university and college admission service that shows the outcomes for a random selection of students who applied to that university over the past 5 years.
- A set of data collected by the BBC that shows the outcomes for all of their journalists from various backgrounds who went to that university over the past 5 years.
- A set of data collected by the Girls' Day School Trust that shows the outcomes for all their students who went to that university over the past 20 years.
- A set of data collected by the School of Computer Science that shows the outcomes for all students who studied for a degree over the past 10 years.

Quiz:
STEM SMART Computer Science Week 30

Advantages and limitations of LLMs

Advanced: Practice 2

Large language models (LLMs) are trained on huge volumes of text and generate responses by predicting which words are most likely to come next. Like any AI tool, they have both advantages and limitations.

For each statement about large language models, drag either **Advantage** or **Limitation** into the correct column.

Statement	Category
An LLM can generate a first draft of a creative writing piece quickly.	
An LLM may not know about events that happened after its training data was last updated.	
A vague or poorly worded prompt is likely to produce a less accurate response.	
An LLM can produce fluent, human-like text that reads naturally.	
An LLM may reflect biases present in the data it was trained on.	
An LLM produces responses that seem reasonable based on its training data, but does not verify whether they are factually correct.	

Items:

Limitation Advantage

Quiz:
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Identifying problems in an LLM response

Advanced: Challenge 1

A student asked a large language model: "Tell me about the history of the internet."

LLM response:

> The internet has its origins in ARPANET, a US military research project launched in 1969. The first message ever sent over this network was a simple 'Hello', transmitted between researchers at UCLA and Stanford University.
>
> In the early 1970s, American computer scientists Vint Cerf and Robert Kahn developed TCP/IP, the set of communication protocols that would become the foundation of the modern internet.
>
> The World Wide Web was later invented by British scientist Tim Berners-Lee in 1991, making the internet accessible to the general public for the first time.
>
> Today, the internet connects billions of people worldwide and, as of this year, there are estimated to be over 5 billion active social media users globally.

Select all of the statements that correctly identify a problem with the LLM response above.

- [ ] The response does not include enough detail about social media to be useful.
- [ ] The response contains a factual error about the first message sent over ARPANET.
- [ ] The response is unreliable because LLMs can only answer questions about computing topics.
- [ ] The response makes a claim about the current number of social media users, but an LLM cannot access up-to-date information.
- [ ] The response presents internet history from a predominantly American perspective, which may reflect bias in the training data.
- [ ] The fluent and confident writing style makes it easy to accept the response as accurate without checking the facts.
- [ ] The response uses language that is too technical for a general audience.

Quiz:
STEM SMART Computer Science Week 30

Social and ethical issues with AI systems

Advanced: Practice 2

LLM marked question

30 attempts remaining today

A company has recently started using an AI system to screen job applications. The AI system analyses applications and selects candidates based on specified criteria set by the company.

Identify two social or ethical impacts of using an AI system in the hiring process.

[2 marks]

Quiz:
STEM SMART Computer Science Week 30

ML types: 2

Advanced: Practice 1

There are three main types of machine learning: supervised, unsupervised, and reinforcement.

For each of the definitions shown below, drag and drop the most appropriate type to the space alongside the definition.

Example	Level
A form of machine learning where the model is trained using trial and error. The model can be said to "learn" from mistakes and will constantly improve.	
A form of machine learning where the model is trained using labelled data. The model is trained to find the patterns, relationships, and features that map each piece of training data to its associated label.	
A form of machine learning where the model is trained on an unlabelled data set. The model must identify patterns, hidden relationships, or structures within the data.	

Items:

Supervised learning Reinforcement learning Unsupervised learning

Quiz:
STEM SMART Computer Science Week 30

ML applications: collaborative filtering

Advanced: Challenge 1

A music streaming service uses a machine learning technique called collaborative filtering to recommend songs to its users.

There are a number of steps that are taken in order to train the machine learning model.

Put the following steps into the correct order to describe how the recommendation system works, starting with the first step.

Available items

- The model is trained to identify patterns in listening habits.
- The system identifies unheard songs enjoyed by similar users.
- The system collects user listening data, including plays and likes.
- The system identifies other users whose listening history is similar to the current user.
- The system recommends the songs most likely to be enjoyed.

Quiz:
STEM SMART Computer Science Week 30

ML models: lifecycle

Advanced: Practice 2

A project lifecycle refers to the series of phases or stages that a project progresses through, from initiation to completion.

Place the stages of developing a machine learning model into a logical order.

Available items

Preparing the data
Training the model
Evaluating the model
Explaining the model
Testing the model
Defining the problem

Quiz:
STEM SMART Computer Science Week 30

ML models: testing and evaluating

Advanced: Practice 2

A machine learning model has been developed to identify birds from their sound alone.

Part A

Whilst testing and evaluating the model, the developers found the following results. Which one of them is an indication of **bias** in the machine learning model?

- The model is unable to identify human speech from sound.
- The app is 96% accurate for pigeons and 48% accurate for blackbirds.
- The average accuracy across all birds is 68%.

Part B

Whilst testing and evaluating the model, it was found that it appeared to be biased against blackbirds. Which **two** of the following actions could the developers take to improve their model?

- Add audio samples of blackbirds in different conditions such as close up or far away.
- Add more audio samples of a wide variety of birds.
- Add audio samples of humans and other animals in a variety of environments.

ML engines: decision trees

Advanced: Practice 2

Decision trees are engines that are a common component of many machine learning models.

A decision tree is made up of nodes that can be described as root, leaf, or decision nodes. Which **two** of the following types of node would contain a condition?

- [ ] Leaf node
- [ ] Root node
- [ ] Decision node

Quiz:
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